



EEE103: Computer Programming
Brac University, Summer 2025
Assignment-01

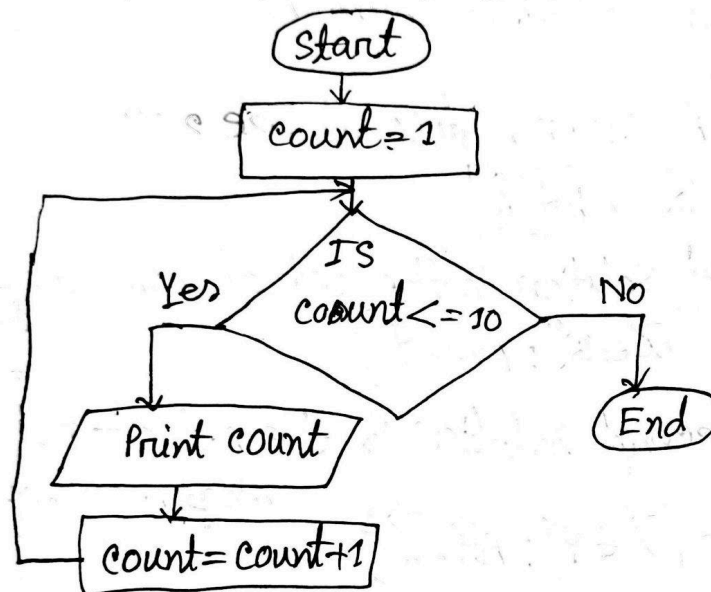
Name: Souvik Barman Ratul
ID: 24121205, Section: 06

1. Draw a flowchart and write a c-programm that prints numbers from 1 to 10.

Flowchart:

Answer to the Q-1

* Draw a flowchart and write a c program that prints number from 1 to 10



Code:

```
#include <stdio.h>
int main() {
    int count=1;
    while (count<=10)
    { printf("%d",count);
      count=count+1;
    }
    return 0;
}
```

Output

12345678910

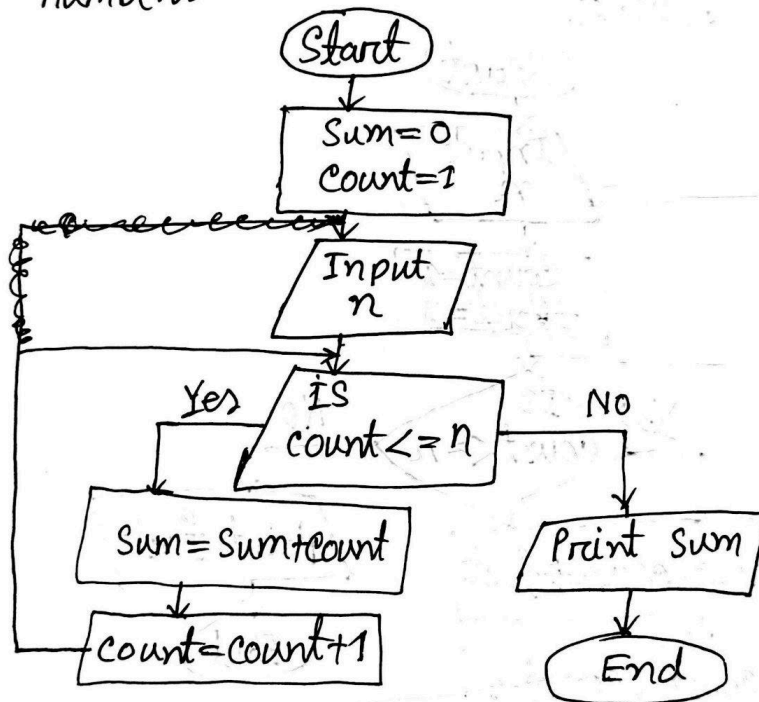
=== Code Execution Successful ===

2. Draw a flowchart and write a c-program that prints the sum of first N Natural Numbers.

Flowchart:

Answer to the Q. No- 2

Draw a flowchart and write a c-Program that prints the sum of first N Natural numbers.



Code:

```
#include <stdio.h>

int main() {
    int n, sum = 0, i;
    printf("Enter a Number ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        sum = sum + i;
    }
    printf("The sum of the first %d natural numbers is %d\n", n, sum);
    return 0;
}
```

Output

```
Enter a Number 5
The sum of the first 5 natural numbers is 15

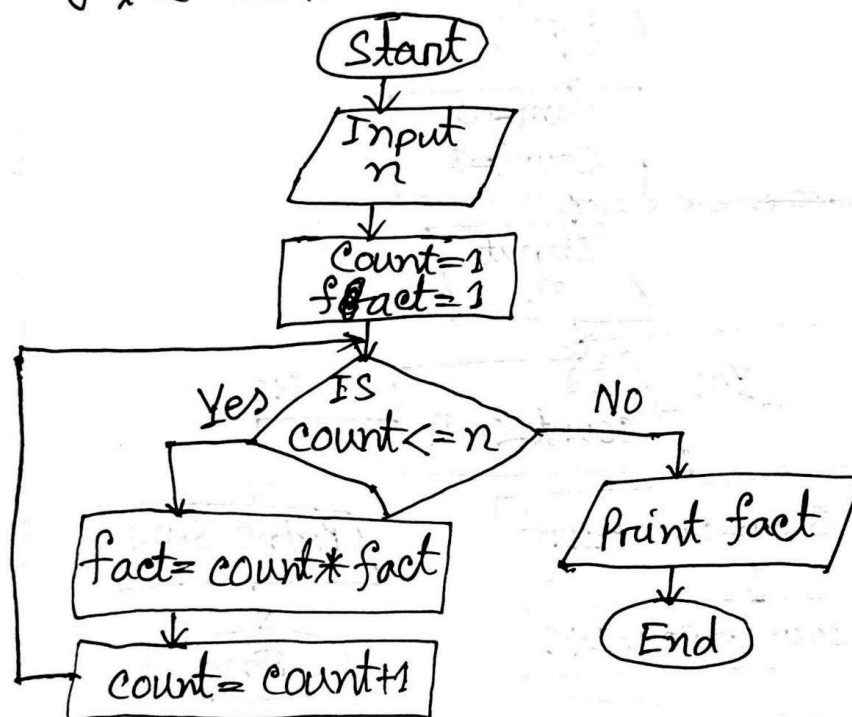
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```

3) Draw a flowchart and write a program to find the factorial of a given number by the user.

Flowchart:

Answer to the Q. NO-3

Draw a flowchart and write a program to find the factorial of given number by the user.



Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i;
```

```
    long long factorial = 1;
```

```
    printf("Enter a Number: ");
```

```
    scanf("%d", &n);
```

```
    if(n < 0) {
```

```
        printf("\n");
```

```
    } else {
```

```
        for(i = 1; i <= n; i++) {
```

```
            factorial = factorial * i;
```

```
        }
```

```
        printf("Factorial of %d is %llu\n", n, factorial);
```

```
    }
```

```
    return 0;
```

```
}
```

Output

```
Enter a Number: 5
```

```
Factorial of 5 is 120
```

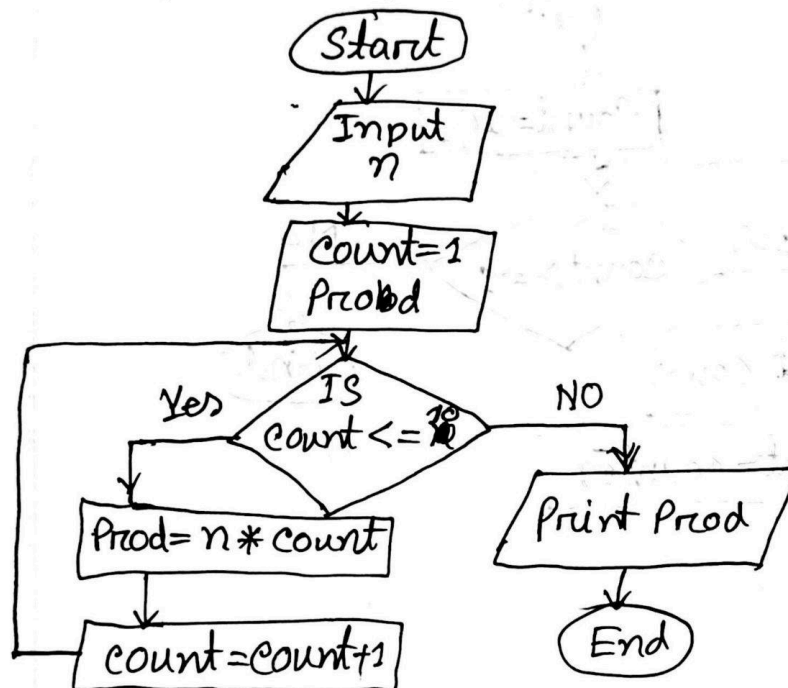
```
=== Code Execution Successful ===
```

4) Draw a flowchart and write a program that takes an integer and prints its multiplication table up to 10.

Flowchart:

Ans to the Q. NO-4

Draw a flowchart and write a program that takes an integer and prints its multiplication table up to 10.



Code:

```
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter an integer ");
    scanf("%d", &n);
    printf("Multiplication table of %d:\n", n);
    for(i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }
    return 0;
}
```

Output

```
Enter an integer 5
Multiplication table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

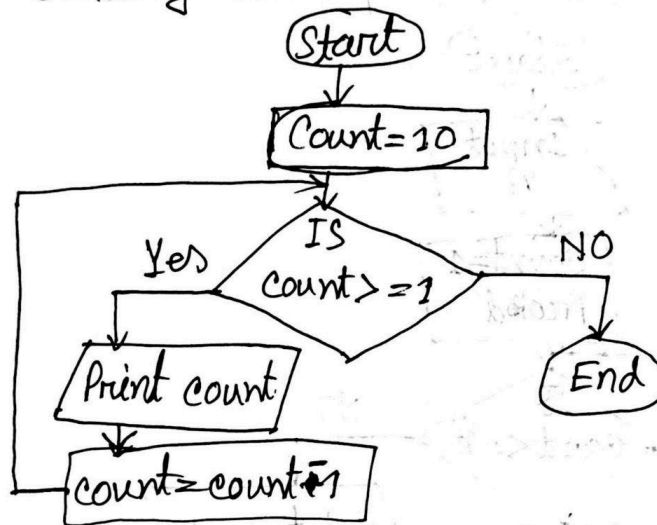
=== Code Execution Successful ===

5) Draw a flowchart and write a program that prints all the numbers from 10 to 1 in descending order.

Flowchart:

Ans to the Q. NO - 5

Draw a flowchart and write a program that prints all the numbers from 10 to 1 in descending order.



Code:

```
#include <stdio.h>
int main() {
    int i;
    for(i=10;i>=1;i--)
    { printf("%d",i);
      if(i>1)
        printf(",");
    }
    return 0;
}
```

Output

10,9,8,7,6,5,4,3,2,1

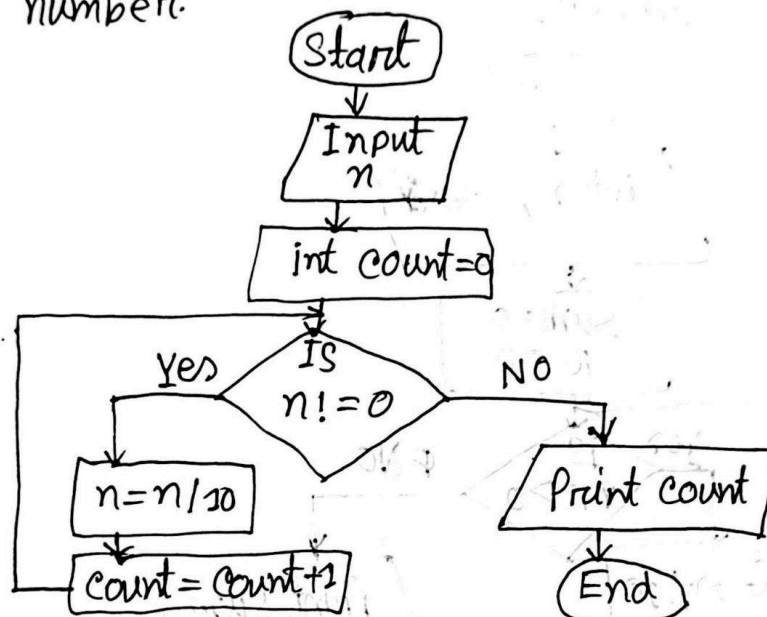
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6) Draw a flowchart and write a program to count the number of digits in a given number.

Flowchart:

Answer to the Q. No-6

Draw a flowchart and write a program to count the number of digit in a given number.



Code:

```
#include <stdio.h>

int main() {
    int num, count = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    if(num == 0) {
        count = 1;
    } else {
        while(num != 0) {
            num = num / 10;
            count++;
        }
    }
    printf("Number of digits: %d\n", count);
    return 0;
}
```

Output

Enter a number: 2890208

Number of digits: 7

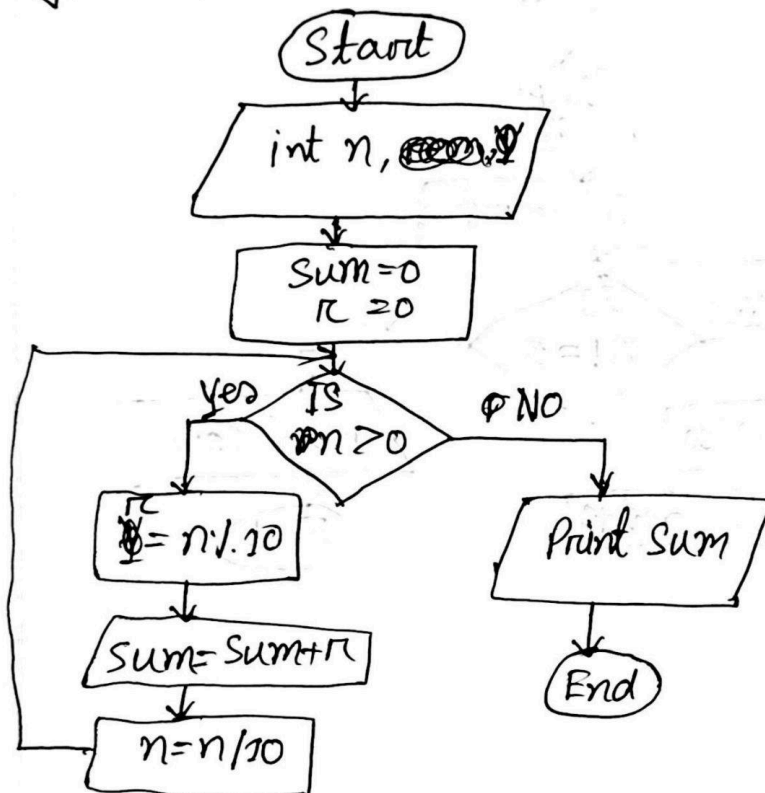
=== Code Execution Successful ===

7) Draw a flowchart and write a program to find the sum of the digits of a given number.

Flowchart:

Ans to the Q. NO-07

Draw a flowchart and write a program to find the sum of the digits of a given number.



Code:

```
#include <stdio.h>
int main() {
    int num, digit, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    while (num != 0) {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }
    printf("Sum of digits: %d\n", sum);
    return 0;
}
```

Output

Enter a number: 18121036

Sum of digits: 22

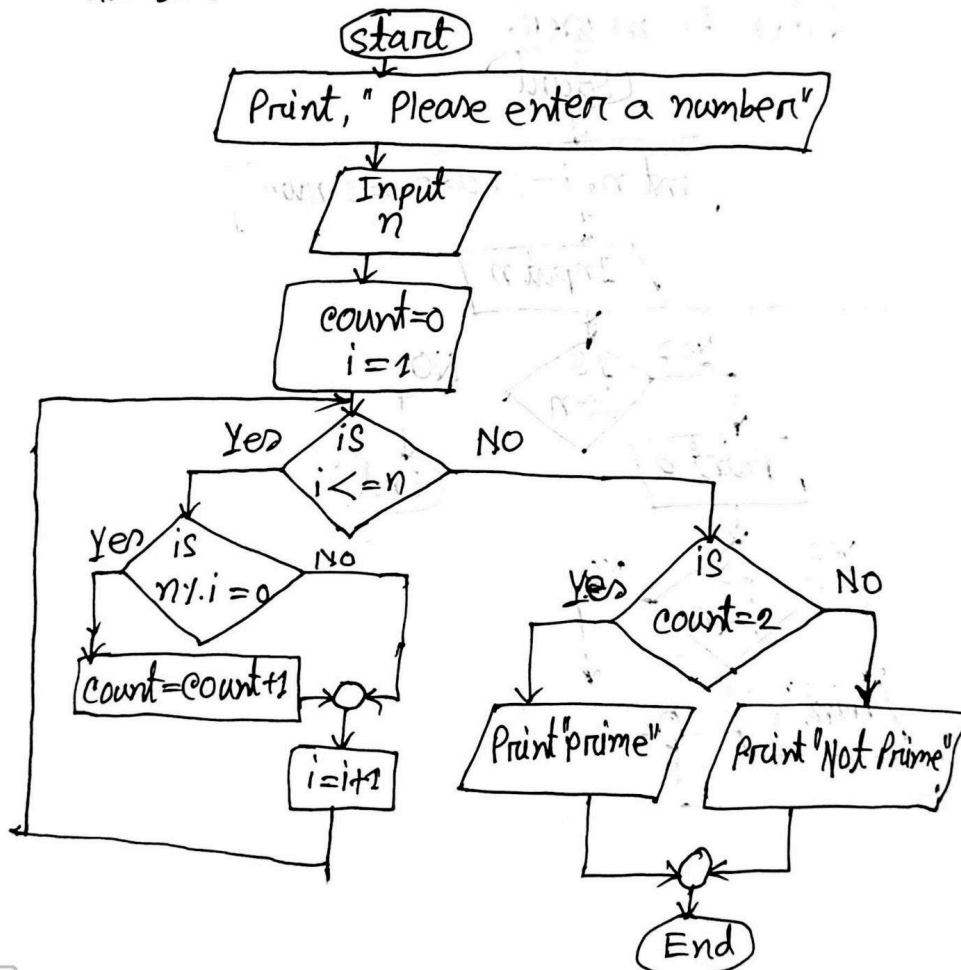
=== Code Execution Successful ===

8) Draw a flowchart and write a program to check whether a given number is prime or not.

Flowchart:

Ans to the Q. No-8

Draw a flowchart and write a program to ~~generate the first 10~~ check whether a given number is prime or not.



Code:

```
#include <stdio.h>

int main() {
    printf("Enter a number");
    int n, count=0, i=1;
    scanf("%d",&n);
    while (i<=n)
    {
        if (n%i==0)
        {
            count=count+1;
        }
        i=i+1;
    }
    if (count==2)
    {
        printf("The number is Prime");
    }
    else{
        printf("The number is Not Prime");
    }
    return 0;
}
```

Output

```
Enter a number5
The number is Prime
```

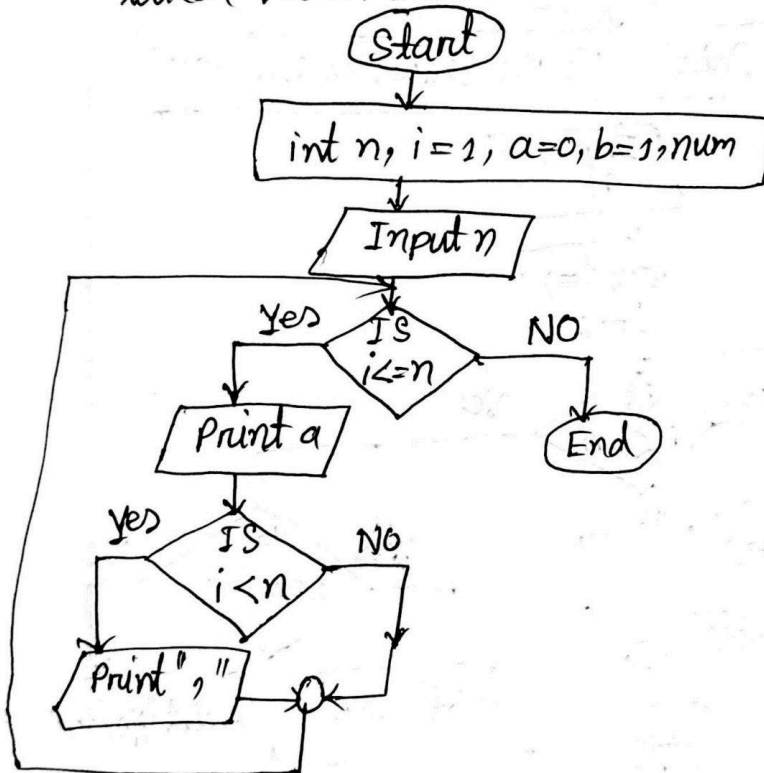
```
=== Code Execution Successful ===
```

9) Draw a flowchart and write a program to generate the first N terms of the Fibonacci series. The value of N must taken from user.

Flowchart:

Ans to the Q. No-9

Draw a flowchart and write a program to generate the first N terms of the Fibonacci Series. The value of N must taken from user.



Code:

```
#include <stdio.h>

int main() {
    int n, i;
    int a = 0, b = 1, num;
    printf("Enter the number of terms: ");
    scanf("%d", &n);
    printf("The first %d terms of the Fibonacci series are:\n", n);
    for (i = 1; i <= n; i++) {
        printf("%d ", a);
        num = a + b;
        a = b;
        b = num;
    }
    return 0;
}
```

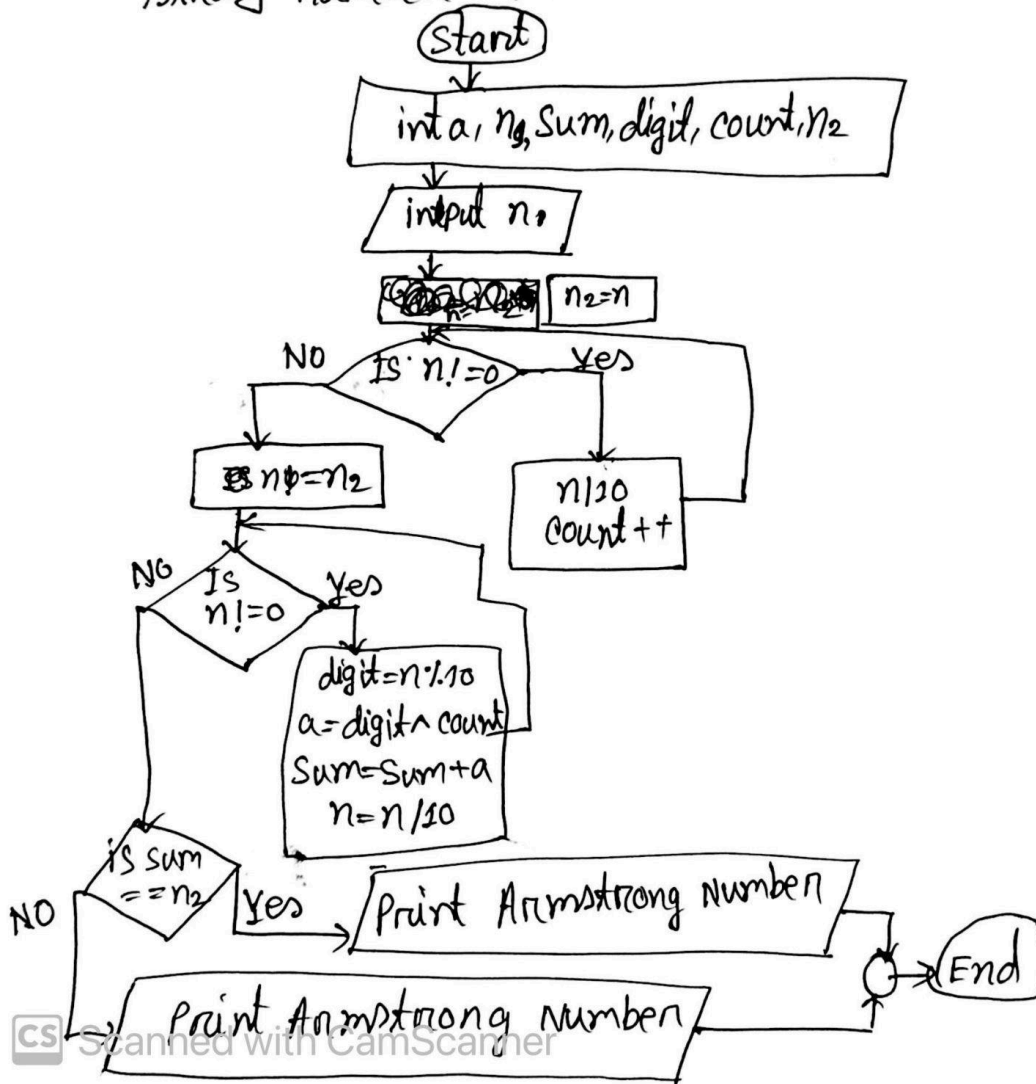
Output

```
Enter the number of terms: 8
The first 8 terms of the Fibonacci series are:
0 1 1 2 3 5 8 13
```

```
=== Code Execution Successful ===
```

Flowchart:

Draw a flowchart and write a program to check whether a given number is an Armstrong number or not.



Code:

```
#include <stdio.h>

int main() {
    int num, originalNum, remainder, result = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    original Num = num;
    while (num != 0) {
        remainder = num % 10;
        result += remainder * remainder * remainder;
        num /= 10;
    }
    if (result == originalNum)
        printf("%d is an Armstrong number.\n", originalNum);
    else
        printf("%d is not an Armstrong number.\n", originalNum);
    return 0;
}
```

Output

```
Enter a number: 153
153 is an Armstrong number.
```

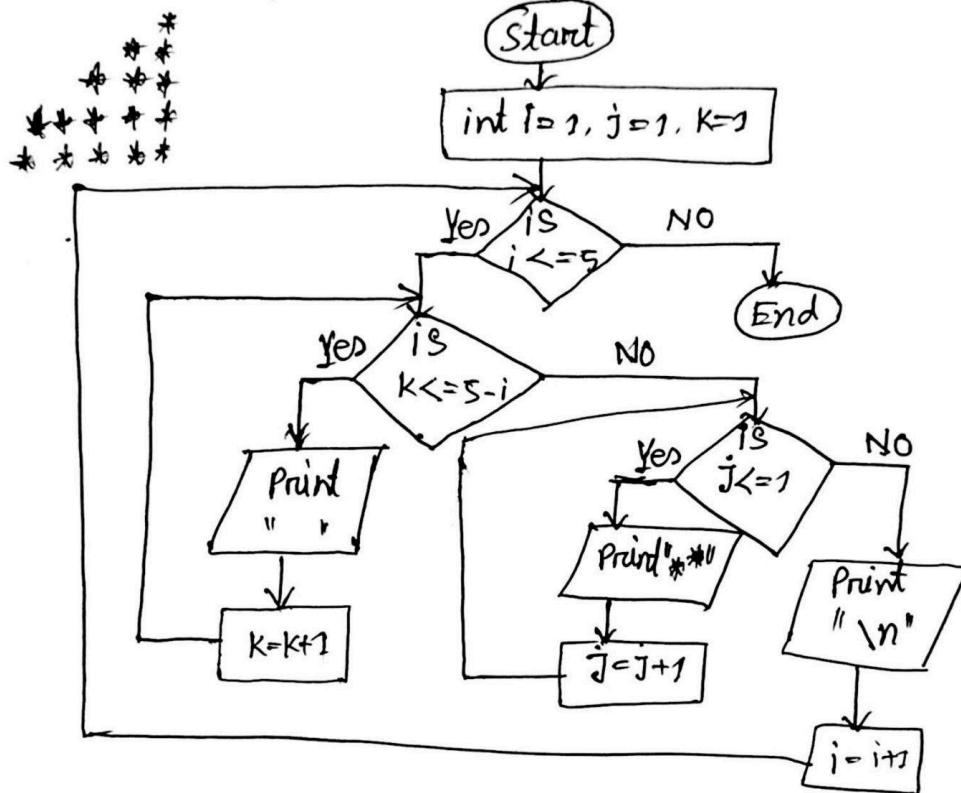
```
=== Code Execution Successful ===
```

11) Draw a flowchart and write a program to print the following pattern using nested for loops

Flowchart:

Ans to the Q. No- 11

Draw a flowchart and write a program to print the following pattern using nested for loops.



Code:

```
#include <stdio.h>

int main() {
    int i,j,k;
    for(i=1;i<=5;i++)
    {
        for(j=i;j<=5;j++)
        {
            printf(" ");
        }
        for(k=1;k<=i;k++)
        {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
  *
  **
 ***
****
*****
```

=== Code Execution Successful ===